

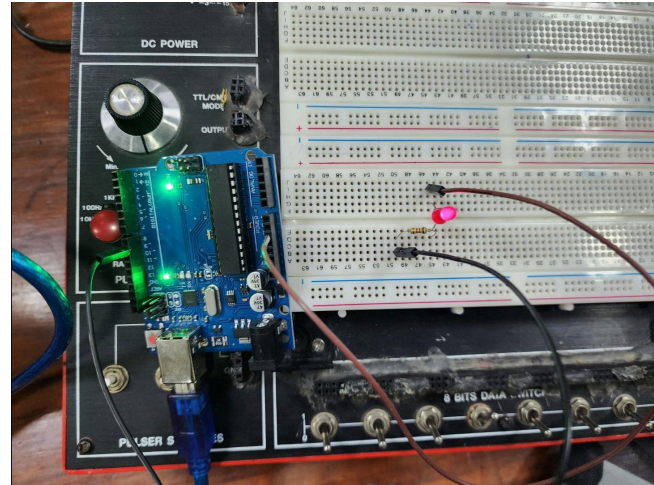
Lab # 3

23L-6012, 23L-6006

Question 1:

```
1 String inputString = "";
2
3 void setup() {
4   Serial.begin(9600);
5   pinMode(13, OUTPUT);
6 }
7
8 void loop() {
9   if (Serial.available() > 0) {
10    inputString = Serial.readStringUntil('\n');
11    inputString.trim();
12
13    if (inputString=="ON"){
14      digitalWrite(13,HIGH);
15      Serial.println("Led is ON.");
16    }
17    else if(inputString=="OFF"){
18      digitalWrite(13,LOW);
19      Serial.println("Led is OFF.");
20    }
21  }
22 }
23 }
```

Led is ON.
Led is OFF.
Led is ON.
Led is OFF.
Led is ON.
Led is OFF.
Led is ON.
Led is OFF.
Led is ON.
Led is OFF.
Led is ON.
Led is OFF.
Led is ON.
Led is OFF.
Led is ON.
Led is OFF.
Led is ON.
Led is OFF.
Led is ON.
Led is OFF.
Led is ON.
Led is OFF.
Led is ON.

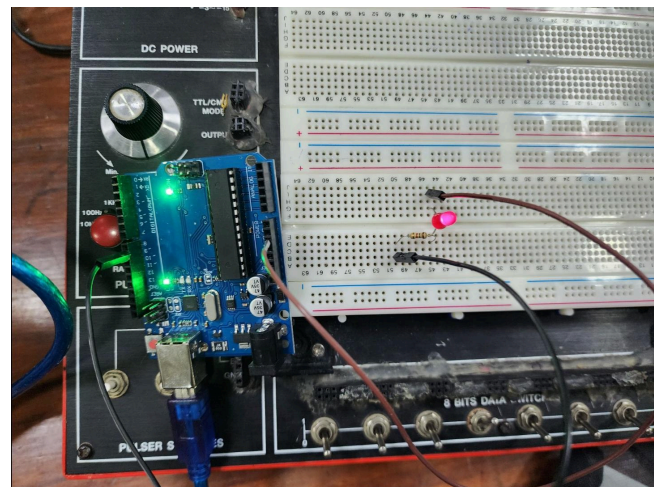


Question 2:

```
1 int ledPin = 9;
2
3
4 void setup() {
5   Serial.begin(9600);
6   pinMode(ledPin, OUTPUT);
7 }
8
9 void loop() {
10  if (Serial.available()) {
11    int brightness = Serial.parseInt();
12
13    int pwmvalue=map(brightness,0,100,0,255);
14    analogWrite(9,pwmvalue);
15    Serial.print("Brightness is: ");
16    Serial.println(brightness);
17    delay(1000);
18  }
19 }
```

Brightness is: 50
Brightness is: 100
Brightness is: 0
Brightness is: 80
Brightness is: 0

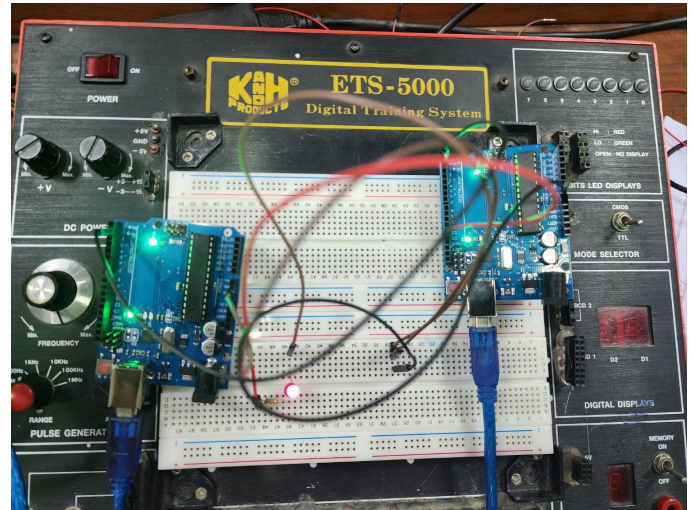
Output Serial Monitor X
Message (Enter to send message to



Question 3:

Slave:

```
sketch_feb12a.ino
1 String message = "";
2
3 void setup() {
4   Serial.begin(9600);
5   pinMode(13, OUTPUT);
6 }
7
8 void loop() {
9   if (Serial.available() > 0) {
10    message = Serial.readStringUntil('\n');
11    message.trim();
12
13    if (message == "ON") {
14      digitalWrite(13, HIGH);
15    }
16    else if (message == "OFF") {
17      digitalWrite(13, LOW);
18    }
19  }
20 }
```



Master:

```
3 void setup() {
4   Serial.begin(9600);
5 }
6 }
7
8 void loop() {
9   if (Serial.available() > 0) {
10    String message = Serial.readStringUntil('\n');
11    message.trim();
12    Serial.println(message);
13  }
14 }
```

OFF
ON
OFF
ON
OFF
ON
OFF
ON
ON
ON
OFF
ON
ON

4. How DC Motor Speed is Controlled using PWM?

Basic Idea:

- Motor connected through Motor Driver (L298N / L293D / MOSFET)
- PWM signal applied to Enable pin

Concept Explanation:

PWM = Pulse Width Modulation

Instead of reducing voltage, we switch voltage ON and OFF very fast.

Average voltage changes → Motor speed changes.

Example:

Duty Cycle Effective Voltage Speed

0% 0V Stop

25% Low Slow

50% Medium Medium

100% Full Voltage Full Speed

Why is PWM Used?

- Efficient (no heat loss like resistor method)
- Better control
- Energy saving
- Used in robotics & IoT systems